

**PRECLINICAL MODULE III:
METABOLIC HOMEOSTASIS AND MOLECULAR GENETICS (MHMG 1102)**

Date	Learning Activity	Topic	Hrs.
Week 1			
Monday April 25, 2022	<i>Morning:</i> Lecture 1 (Dr. Tibebe Antonios)	Introduction to Bioenergetics: The laws of thermodynamics, free energy, the link between redox potential and free energy, high energy molecules and coupled reactions. Biochemical Catalysis and Enzymes: The main features of enzymes, enzyme nomenclature, introduction to cofactors and substrate binding models.	2.5
	<i>Afternoon</i>		
Tuesday April 26, 2022	<i>Morning:</i> Lecture 2 (Dr. Tibebe Antonios)	Mechanisms of enzyme catalysis, enzyme kinetics with emphasis on the importance of V_{max} and K_m ; the effects of pH and temperature; enzyme inhibition	2.5
	<i>Afternoon</i>		
Wednesday April 27, 2022	<i>Morning:</i> Lecture 3 (Dr. Tibebe Antonios)	Enzyme regulation and the clinical application of enzymes Coenzymes, cofactors and their functions	2.5
	<i>Afternoon:</i>		
Thursday April 28, 2022	<i>Morning:</i> Lecture 4 (Muluken Fekadie)	Carbohydrate Metabolism: Enzymes and transporters involved in the digestion and absorption of carbohydrates, the importance of glycolysis in different cells and tissues and glycolytic reactions	2
	<i>Afternoon</i>		
Friday April 29, 2022	<i>Morning:</i> Lecture 5 (Muluken Fekadie)	Regulation of glycolysis and feeder pathways, importance of the pentose phosphate pathway: different cells and tissues and its reactions	2.5
	<i>Afternoon:</i>		
		Weekly contact hour	12

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Week 2			
Date		Topics	Hrs
Monday May 2, 2022	<i>Morning:</i> Lecture 6 (Muluken Fekadie)	Regulation of the pentose phosphate pathway; the glucuronic acid pathway and its functions; glycogen storage sites, synthesis and degradation; regulation of glycogen metabolism	3
	<i>Afternoon</i>	Lab session I : Determination of ALT, AST and ALP (Groups 1, 2, 5 & 6)	2
Tuesday May 3, 2022	<i>Morning:</i> Lecture 7 (Muluken Fekadie)	The significance of gluconeogenesis and responsible organs, reactions and regulation and the Cori cycle. The Pyruvate Dehydrogenase Reaction TCA Cycle and its regulations	2.5
	<i>Afternoon:</i>	Lab session I : Determination of ALT, AST and ALP (Groups 3 & 4)	2
Wednesday May 4, 2022	<i>Morning:</i> Lecture 8 (Mohammed Mehdi)	Lipid Metabolism: Lipid digestive enzymes and bile salts, transport of triacylglycerols to the adipose, triacylglycerol mobilization; entry of fatty acids into the cell and the mitochondria Fatty Acid Oxidation: β -oxidation reactions, energy yield	3
	<i>Afternoon</i>	Lab session II: Blood glucose estimation (Groups 1, 2, 5 & 6)	2
Thursday May 5, 2022		Holiday	
Friday May 6, 2022	<i>Morning:</i> Lecture 9 (Mohammed Mehdi)	Regulation of β -oxidation, oxidation of odd and unsaturated fatty acids; Minor pathways of fatty acid oxidation; The synthesis and utilization of ketone bodies Fatty acid biosynthesis: overview, short term and long term regulation; the synthesis of triacylglycerols	3
	<i>Afternoon</i>	Seminar I	3
		Weekly contact hour	20.5

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Week 3			
Date		Topics	Hrs
Monday May 9, 2022	<i>Morning: Test 1</i>	Written Exam-1 (lectures 1-8 and seminars 1)	0.75
	<i>Afternoon</i>	Lab Session II Cont'd: Blood glucose estimation (Groups 3 & 4)	2
Tuesday May 10, 2022	<i>Morning: Lecture10 (Mohammed Mehdi)</i>	The synthesis and degradation of glycerophospholipids and sphingolipids. Classes of eicosanoids and their synthesis Cholesterol metabolism: synthesis, regulation, derivatives of cholesterol; lipoproteins and the transport of cholesterol	3
	<i>Afternoon</i>	Lab Session III Determination of TAG, HDL, LDL (Lipid Profile) (Groups 1, 2, 3, & 4)	2
Wednesday May 11, 2022	<i>Morning: Lecture 12 (Maria)</i>	Amino Acid Metabolism: The activation and specificity of proteolytic enzymes; absorption of amino acids; protein turnover: lysosomal and the ubiquitin/proteasome pathway; the removal and transport of amino groups from amino acids,	2.5
	<i>Afternoon</i>	Lab Session III Cont'd: Determination of TAG, HDL, LDL (Lipid Profile) (Groups 5 & 6)	2
Thursday May 12, 2022	<i>Morning: Lecture 13 (Maria)</i>	The urea cycle: reactions, regulation, salvage. The fate of carbon skeletons of amino acids. One carbon metabolism: sources of one carbon; coenzymes involved. Special products synthesized from amino acids	3
	<i>Afternoon</i>	Lab Session IV: Renal function tests: urea and creatinine (Groups 3, 4, 5 & 6)	2
Friday May 13, 2022	<i>Morning: Lecture 14 (Maria)</i>	Inborn errors in amino acid metabolism	2
	<i>Afternoon</i>	Lab Session IV Cont.: Renal function tests: urea and creatinine (Groups 1 & 2)	2
		Weekly contact hours	21.25

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Week 4			
Monday May 16, 2022	<i>Morning: Lecture 11 (Tinsae G)</i>	<i>Electron Transport and Oxidative Phosphorylation:</i> Components of the respiratory chain, coupling of electron transport and oxidative phosphorylation, the energy yield of reduced coenzymes, inhibitors and uncouplers	3
	<i>Afternoon</i>	Seminar II	3
Tuesday May 17, 2022	<i>Morning: Lecture 15 (Tinsae G)</i>	Xenobiotic metabolism	2
	<i>Afternoon</i>	Seminar II (Cont'd)	3
Wednesday May 18, 2022	<i>Morning: Lecture 17 (Dr Tsehayneh Kelemu)</i>	<i>Nucleotide Metabolism:</i> Purine metabolism: <i>de novo</i> and salvage pathways, degradation; pyrimidine metabolism; deoxyribonucleotide synthesis	2.5
	<i>Afternoon</i>		
Thursday May 19, 2022	<i>Morning: Lecture 18 (Dr. Tsehayneh Kelemu)</i>	<i>Molecular Genetics:</i> The packaging of DNA in chromosomes, the nucleoid; Prokaryotic and eukaryotic DNA replication	3
	<i>Afternoon:</i>		
Friday May 20, 2022	<i>Morning: Lecture 19 (Dr Tsehayneh Kelemu)</i>	Mutations and DNA repair mechanisms; Prokaryotic and eukaryotic gene transcription	3
	<i>Afternoon</i>		
		Weekly contact hour	19.5

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Week 5			
Monday May 23, 2022	<i>Morning: Test 2</i>	Written Exam-2 from lectures 9-16	0.75
	<i>Afternoon</i>		
Tuesday May 24, 2022	<i>Morning: Lecture 20 (Dr. Tsehayneh Kelemu)</i>	Translation: tRNA structure, types of ribosomes, the genetic code, stages of translation and post-translational modifications.	2
	<i>Afternoon</i>		
Wednesday May 25, 2022	<i>Morning: Lecture 21 (Dr. Tsehayneh Kelemu)</i>	Regulation of prokaryotic and eukaryotic genetic expressions	2
	<i>Afternoon:</i>		
Thursday May 26, 2022	<i>Morning: Lecture 22 (Dr Tsehayneh Kelemu)</i>	Recombinant DNA and cloning, gene therapy, DNA amplification, DNA fingerprinting	2
	<i>Afternoon</i>		
Friday May 27, 2022	<i>Morning: (TBA) Lecture 23</i>	Clinical Visit (Diabetic clinic)	2
	<i>Afternoon</i>	Lab quiz: Questions from lab sessions I, II, III and IV	0.75
		Weekly contact hours	9.5

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Week 6			
Monday- Thursday May 30 – June 2, 2022		Study break	
Friday June 3, 2022	<i>Morning:</i> Final examination	Written exam-3 (from all lectures and seminars)	3
Weekly contact hours			3